

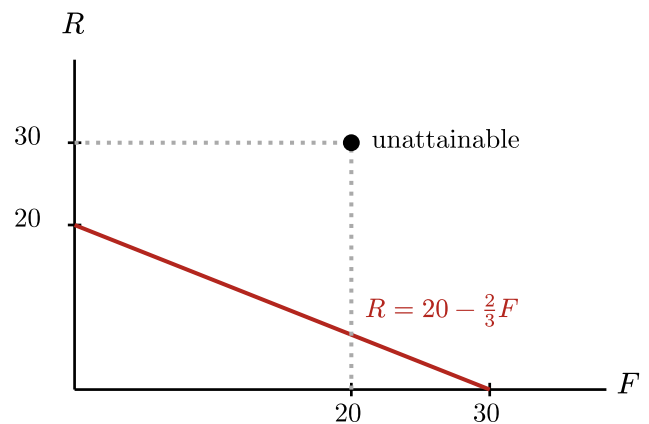
ECON 0100 | Exercise A1 | PPF

I Solutions

Answers and work are shown in the red blocks.

One of Hagrid's unknown skills is that he's great in the kitchen. He can bake 20 rock cakes (R) or 30 fruitcakes (F) in one day. Set up Hagrid's PPF on an x, y graph with rock cakes (R) on the vertical and fruitcakes (F) on the horizontal.

Plot. A full day on rock cakes is the vertical intercept, 20 R , and a full day on fruitcakes is the horizontal intercept, 30 F . The PPF is the straight line between them, $R = 20 - \frac{2}{3}F$. The Q2 bundle (20, 30) is marked above it.



Q1 | Hagrid's PPF

What is Hagrid's opportunity cost of each good?

Opportunity cost of 1R: _____

Opportunity cost of 1F: _____

1R: $\frac{3}{2}$ F. 1F: $\frac{2}{3}$ R.

Hagrid's day is 20R or 30F, so $20R = 30F$. Divide through for each good:

$$1R = \frac{30}{20}F = \frac{3}{2}F \qquad 1F = \frac{20}{30}R = \frac{2}{3}R$$

The two opportunity costs are reciprocals, and $\frac{2}{3}$ is the absolute slope of the PPF: rock cakes given up per extra fruitcake.

Q2 | Feasibility

Hagrid wants to bake 30R AND 20F in one day. Is this inefficient, efficient, or unattainable? Use a graph or algebra to justify your answer.

Inefficient, Efficient, Unattainable: _____

Unattainable.

Graph: the point (20,30) lies outside the PPF, above and to the right of the line, so it is unattainable. At $F = 20$ the frontier only allows $R = 20 - \frac{2}{3} \cdot 20 = \frac{20}{3}$, well short of 30.

Algebra (time check): 30 rock cakes alone take $\frac{30}{20} = \frac{3}{2}$ days and 20 fruitcakes take $\frac{20}{30} = \frac{2}{3}$ of a day, so the bundle needs $\frac{3}{2} + \frac{2}{3} = \frac{13}{6}$ days. Under a full day is inefficient, exactly one day efficient, over one day unattainable. Either justification earns full credit.

Q3 | Dynamics

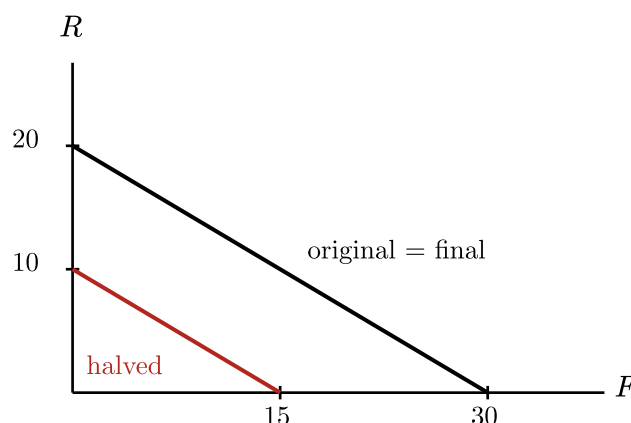
It turns out Hagrid wants to slow down his baking work, and cuts his time in half. Right after, the baking industry goes through a minor revolution and improves the efficiency of everyone's baking by a factor of 2. Show both changes to his PPF.

After both changes Hagrid is back to 20R or 30F a day.

When Hagrid cuts back his hours, his PPF shifts in so that the intercepts are half what they were before: 10R or 15F.

When the new technology arrives, his PPF shifts out to twice what it was prior to the innovation: 20R or 30F. This just so happens to be exactly the same as prior to his time cutback.

Each change scales both intercepts by the same factor, so the slope never moves: his opportunity costs stay at $1R = \frac{3}{2}F$ and $1F = \frac{2}{3}R$ throughout.



Q4 | Which concept (if any) from this block would you want explained in more detail in lecture or recitation?

Completion only: any answer earns the point on Gradescope. Bring the concepts named here to recitation.